

Homework 3: Propensity Score Matching Analysis

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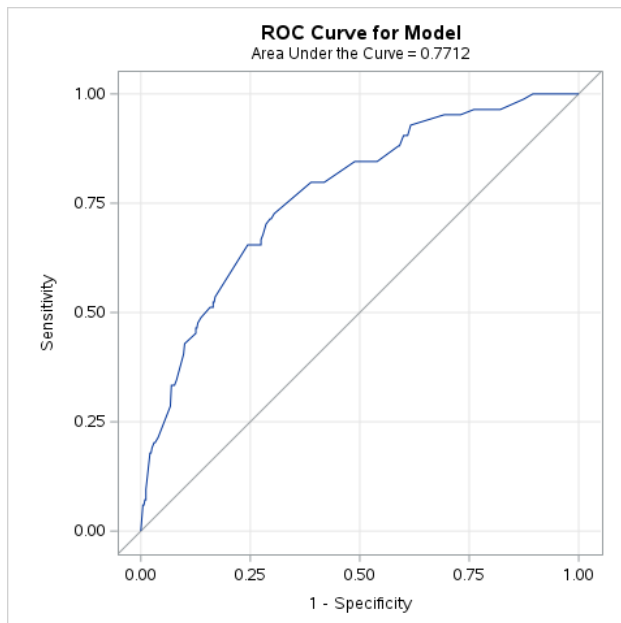
1 Requirement 1: PS Step 1 Model — Model Selection

1.1 Model Specification

The propensity score is defined as the conditional probability of having IVH given baseline covariates: pneumothorax (pneumo), multiple gestation (twin), mode of delivery (Delivery_new), birth location (Inout_new), birth weight category (bwt_cat), and gestational age category (gest_cat). Two logistic regression models were fitted:

- **Model A (No Interactions):** Main effects only.
- **Model B (Two-Way Interactions, Backward):** All two-way interaction terms, backward stepwise at SLS = 0.20.

1.2 ROC Curves



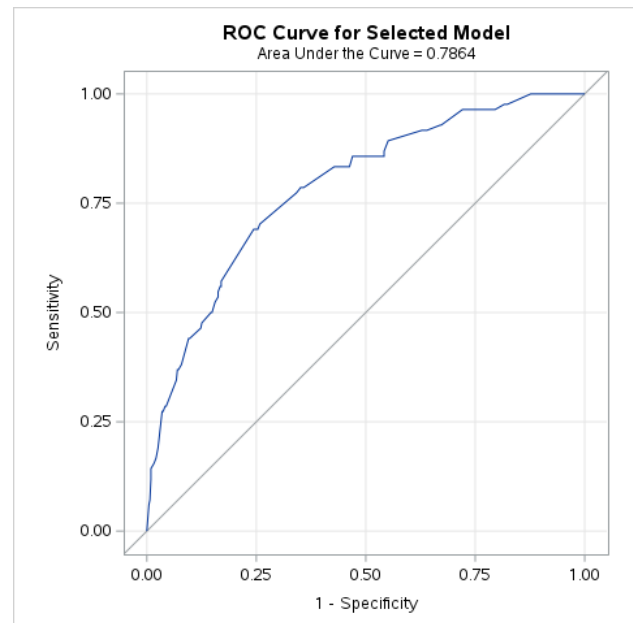
Model A: No Interaction Terms

C-statistic = 0.771

N of matched cohort = 164 (82 pairs)

Hosmer-Lemeshow p-value ≥ 0.05

Quasi-complete separation: No



Model B: Two-Way Interactions (Backward, SLS=0.20)

C-statistic = 0.786

N of matched cohort = 164 (82 pairs)

Hosmer-Lemeshow: not available (selection model)

WARNING: Quasi-complete separation detected

1.3 Model Comparison and Final Model Selection

Criterion	Model A (No Interactions)	Model B (Interactions, Backward)
C-statistic (AUC)	0.771	0.786
Hosmer-Lemeshow p-value	≥ 0.05 (good fit)	Not available
Quasi-complete separation	No	Yes (warning issued)
Matched cohort size	164 (82 pairs)	164 (82 pairs)

Criterion	Model A (No Interactions)	Model B (Interactions, Backward)
Max SMD after matching (binary vars)	0.151	0.151

The following table shows the standardized mean differences (SMD) for each covariate in the matched cohort under Model A:

Obs	Variable	Std Difference (Model A Matched)
1	Logit Propensity Score	0.0105
2	pneumo_yes	-0.0558
3	twin_yes	0.0342
4	delivery_vag	0.0000
5	inout_transport	0.0578
6	bwt_lt1000	-0.1252
7	bwt_1000_1299	0.1512
8	gest_lt28	-0.0259
9	gest_28_31	0.0492

Decision: Model A (no interactions) was selected as the final propensity score model. Although Model B has a marginally higher AUC (0.786 vs. 0.771), it produced quasi-complete separation warnings due to small cell sizes in interaction terms relative to the limited sample size ($N = 514$), indicating numerical instability. Both models produced the same matched cohort size and equivalent balance (all $SMD < 0.2$). For small samples, balance is considered acceptable if $SMD < 0.2$; Model A satisfies this criterion for all covariates with greater numerical stability.

1.4 Final Model A: Parameter Estimates

Covariate	Level	Estimate	OR	95% CI	p-value
Intercept		-4.4905	—	—	< 0.001
Pneumothorax	Yes vs. No	1.3116	3.712	(2.073, 6.648)	< 0.001
Multiple Gestation	Yes vs. No	-1.1410	0.319	(0.136, 0.751)	0.009
Mode of Delivery	Vaginal vs. C-Section	0.7784	2.178	(1.234, 3.844)	0.007
Birth Location	Transport vs. Duke	0.9445	2.571	(1.458, 4.535)	0.001
Birth Weight	< 1000g vs. ≥ 1300 g	0.6698	1.954	(0.785, 4.861)	0.150
Birth Weight	1000–1299g vs. ≥ 1300 g	0.4472	1.564	(0.711, 3.440)	0.266
Gestational Age	< 28 wks vs. ≥ 32 wks	1.6278	5.093	(1.028, 25.221)	0.046
Gestational Age	28–31 wks vs. ≥ 32 wks	1.6220	5.063	(1.145, 22.400)	0.033

C-statistic = 0.771. Pneumothorax, birth location, gestational age, and mode of delivery were significant independent predictors of IVH.

2 Requirement 2: Table 1 — Baseline Characteristics Before and After Matching

Greedy 1:1 nearest-neighbor matching was performed using PROC PSMATCH with a caliper of 0.25 on the logit of the propensity score (Model A). Of 514 infants in the original cohort (84 IVH-exposed; 430 unexposed), **164 infants (82 per group) were successfully matched**, using 97.6% of IVH-exposed infants.

2.1 Covariate Balance — Love Plot

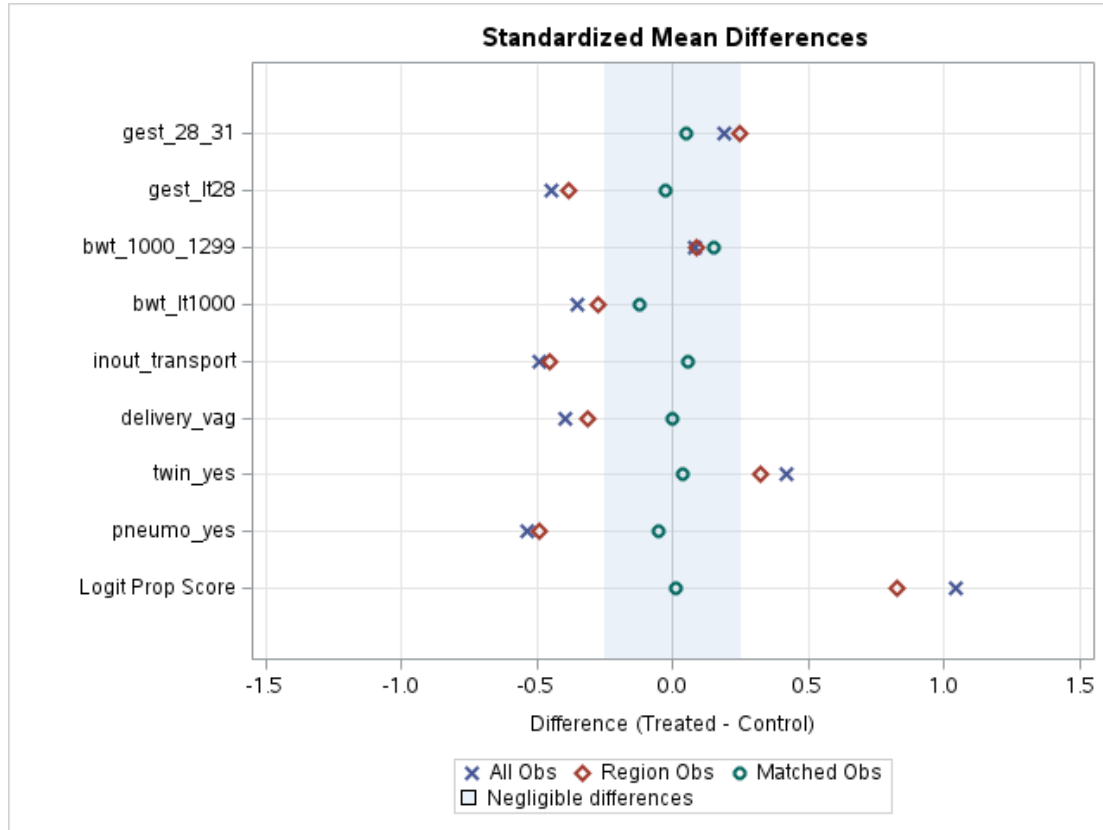


Figure 1: Standardized Difference Plot (Love Plot)

Standardized mean differences for all covariates before matching (blue/triangle) and after 1:1 PS matching (red/circle). All post-matching SMDs fall below the 0.2 threshold, confirming adequate covariate balance.

2.2 Table 1: Baseline Characteristics by IVH Status — Original Cohort (N = 514)

Covariate	No IVH (N = 430)	IVH (N = 84)	p-value	SMD
Pneumothorax			< 0.001	—
No	357 (83.0%)	50 (59.5%)		
Yes	73 (17.0%)	34 (40.5%)		
Multiple Gestation			0.002	—
Singleton	330 (76.7%)	77 (91.7%)		
Multiple	100 (23.3%)	7 (8.3%)		
Mode of Delivery			0.001	—
C-Section	221 (51.4%)	27 (32.1%)		
Vaginal	209 (48.6%)	57 (67.9%)		
Birth Location			< 0.001	—
Born at Duke	366 (85.1%)	54 (64.3%)		
Transport	64 (14.9%)	30 (35.7%)		
Birth Weight Category			0.004	—
< 1000 g	146 (34.0%)	43 (51.2%)		
1000–1299 g	170 (39.5%)	30 (35.7%)		

Covariate	No IVH (N = 430)	IVH (N = 84)	p-value	SMD
1300–1599 g	114 (26.5%)	11 (13.1%)		
Gestational Age Category			< 0.001	—
< 28 weeks	114 (26.5%)	40 (47.6%)		
28–31 weeks	255 (59.3%)	42 (50.0%)		
≥ 32 weeks	61 (14.2%)	2 (2.4%)		

Column percentages shown. Chi-square p-values. In the original cohort, IVH-exposed infants differed significantly from unexposed infants across all six covariates, confirming substantial baseline confounding. IVH-exposed infants were more likely to have pneumothorax (40.5% vs. 17.0%), be transported in (35.7% vs. 14.9%), have lower gestational age (< 28 wks: 47.6% vs. 26.5%), and lower birth weight (< 1000g: 51.2% vs. 34.0%).

2.3 Table 1: Baseline Characteristics by IVH Status — PS Matched Cohort (N = 164)

Covariate	No IVH (N = 82)	IVH (N = 82)	p-value	SMD
Pneumothorax			0.747	-0.056
No	52 (63.4%)	50 (61.0%)		
Yes	30 (36.6%)	32 (39.0%)		
Multiple Gestation			0.787	0.034
Singleton	74 (90.2%)	75 (91.5%)		
Multiple	8 (9.8%)	7 (8.5%)		
Mode of Delivery			1.000	0.000
C-Section	27 (32.9%)	27 (32.9%)		
Vaginal	55 (67.1%)	55 (67.1%)		
Birth Location			0.744	0.058
Born at Duke	52 (63.4%)	54 (65.9%)		
Transport	30 (36.6%)	28 (34.1%)		
Birth Weight Category			0.629	—
< 1000 g	37 (45.1%)	42 (51.2%)		
1000–1299 g	35 (42.7%)	29 (35.4%)		
1300–1599 g	10 (12.2%)	11 (13.4%)		
Gestational Age Category			0.821	—
< 28 weeks	38 (46.3%)	39 (47.6%)		
28–31 weeks	43 (52.4%)	41 (50.0%)		
≥ 32 weeks	1 (1.2%)	2 (2.4%)		

Column percentages shown. After 1:1 PS matching, all p-values exceed 0.62 and all SMDs are well below 0.2 (maximum binary-variable SMD = 0.151 for bwt_1000_1299), indicating adequate covariate balance.

Plain-language interpretation of Table 1: Before matching, IVH-exposed infants were born earlier, weighed less, were more likely to have pneumothorax, and were more frequently transported in from other hospitals — all characteristics reflecting the typical clinical profile of severe prematurity. These imbalances mean any simple comparison of death rates between IVH and non-IVH groups would conflate the effect of IVH with the effect of being more premature. After propensity score matching, both groups look essentially identical on all measured characteristics (all p-values > 0.6), confirming that the matched comparison is valid and free from the measured confounders.

3 Requirement 3: Propensity Score Distribution Before and After Matching

3.1 PS Distribution — Original Cohort (Before Matching)

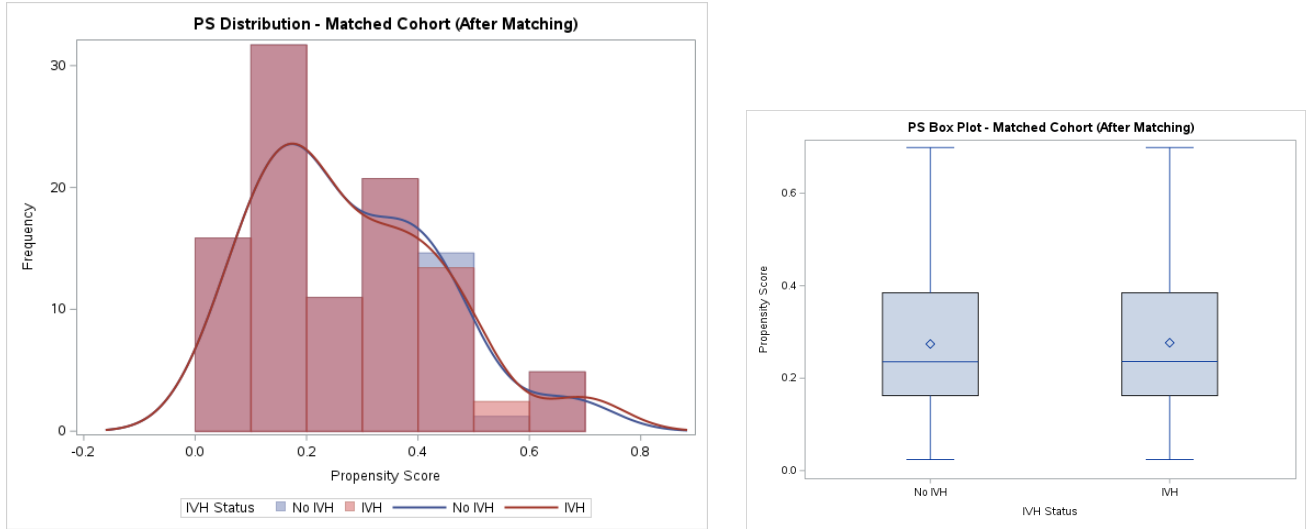


Figure 2: PS Distribution — Original Cohort (N = 514)

Mean PS: IVH = 0.286, No IVH = 0.140. The two groups' PS distributions are clearly separated, reflecting substantial baseline imbalance before matching.

3.2 PS Distribution — PS Matched Cohort (After Matching)

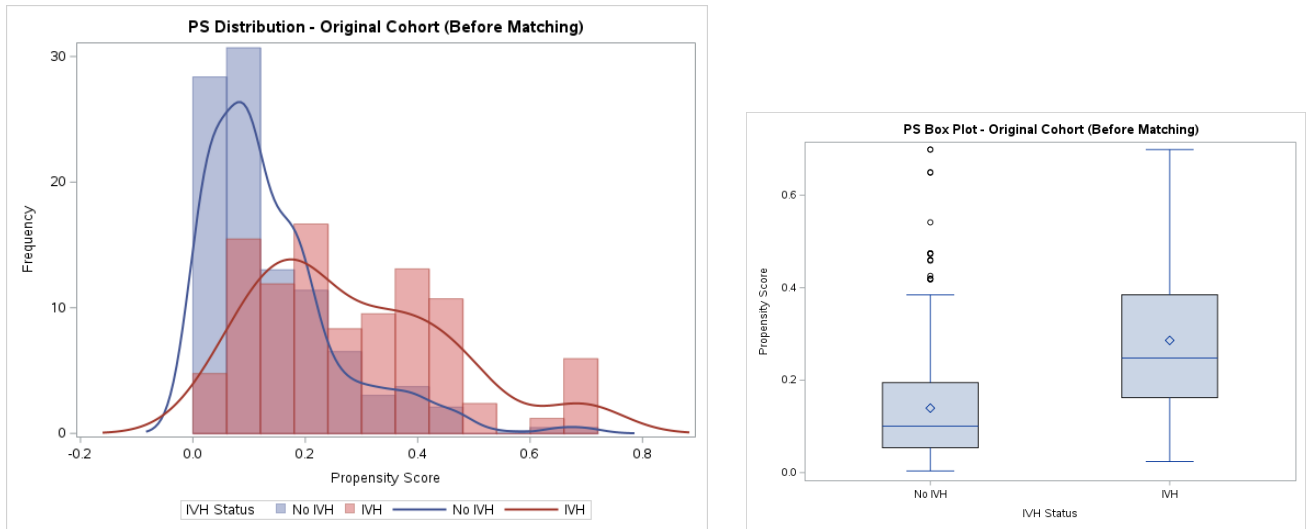


Figure 3: PS Distribution — PS Matched Cohort (N = 164, 82 pairs)

Mean PS: IVH = 0.277, No IVH = 0.274. After matching, the two distributions are nearly identical, confirming successful balance within the region of common support.

Plain-language interpretation: Before matching, IVH-exposed infants had much higher propensity scores (mean 0.286) than unexposed infants (mean 0.140), reflecting their systematically higher baseline risk driven by prematurity. The histograms show largely non-overlapping distributions. After greedy 1:1 matching with a

caliper of 0.25 on the logit PS, both groups had nearly identical PS distributions (mean IVH = 0.277 vs. No IVH = 0.274). The overlapping histograms and aligned box plots confirm that matched controls were truly comparable to IVH-exposed cases in terms of their predicted probability of having IVH.

4 Requirement 4: Table 2 — Unadjusted, Adjusted, and PS-Matched Estimates

Three approaches were used to estimate the association between IVH and mortality:

1. **Unadjusted:** Simple logistic regression (OR) and Poisson regression (RR) with IVH as the only predictor.
2. **Adjusted (HW1 Model):** Multivariable logistic and Poisson regression with IVH + pneumothorax + bwt_cat + gest_cat (purposeful selection from HW1).
3. **PS-Matched:** Conditional logistic regression stratified on matched pair ID (for OR); GEE Poisson regression with matched pair clustering (for RR).

4.1 Table 2: IVH and Mortality — Distribution and Effect Estimates

Mortality distribution by IVH status (row %):

	Dead = 0 (Alive)	Dead = 1 (Dead)
No IVH (N = 430)	372 (86.5%)	58 (13.5%)
IVH (N = 84)	43 (51.2%)	41 (48.8%)

Association estimates (IVH vs. No IVH):

Method	OR (95% CI)	p-value	RR (95% CI)	p-value
Unadjusted	6.116 (3.674, 10.179)	< 0.001	3.619 (2.426, 5.398)	< 0.001
Adjusted (HW1)	4.149 (2.291, 7.516)	< 0.001	2.059 (1.351, 3.137)	< 0.001
PS-Matched	3.857 (1.680, 8.857)	0.002	2.053 (1.349, 3.124)	< 0.001

Adjusted (HW1): multivariable model including pneumothorax, birth weight category, and gestational age category. PS-Matched OR: conditional logistic regression stratified on MatchID. PS-Matched RR: GEE Poisson regression with MatchID as clustering unit.

Plain-language interpretation of Table 2:

- **Unadjusted:** IVH-exposed infants had 6.1 times the odds of dying (OR = 6.116) and 3.6 times the risk of dying (RR = 3.619) compared to unexposed infants. This large crude estimate reflects both the true effect of IVH and the additional risk due to confounders such as extreme prematurity. In raw numbers, 48.8% of IVH-exposed infants died vs. 13.5% of unexposed infants.
- **Multivariable-adjusted (HW1):** After accounting for pneumothorax, birth weight, and gestational age through conventional regression, the adjusted OR fell to 4.149 (95% CI: 2.291–7.516) and the adjusted RR to 2.059 (95% CI: 1.351–3.137). The substantial attenuation from the crude estimate confirms that prematurity-related factors were important confounders.
- **PS-matched:** After matching on the propensity score to create balanced comparison groups, the estimated OR was 3.857 (95% CI: 1.680–8.857) and the estimated RR was 2.053 (95% CI: 1.349–3.124). These are highly consistent with the HW1 adjusted estimates, providing mutual validation across two methodologically distinct approaches.

- **Preferred measure:** The Risk Ratio is preferred over the Odds Ratio because the outcome is common (overall mortality = 19.3%; among IVH-exposed = 48.8%), making the OR substantially overestimate the true relative risk under the rare disease assumption. The PS-matched RR of **2.053** is the most appropriate summary measure.
 - **Conclusion:** After controlling for baseline confounding through propensity score matching, IVH was independently associated with approximately **2-fold higher risk of death** (PS-matched RR = 2.053, 95% CI: 1.349–3.124, $p < 0.001$) in VLBW infants. This finding is robust: both conventional multivariable regression and PS matching — using different statistical assumptions — converge on the same conclusion. The consistency of results strengthens the evidence that IVH is an independent risk factor for mortality, not merely a marker of prematurity.
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5 Summary

Propensity score matching using Model A (main effects only, C-statistic = 0.771) successfully balanced all six baseline covariates between IVH-exposed and unexposed infants (all matched SMD < 0.16), yielding a matched cohort of 164 infants (82 pairs). PS distributions before and after matching confirmed the shift from clearly separated to overlapping distributions, validating the matching procedure. The PS-matched risk ratio (RR = 2.053, 95% CI: 1.349–3.124) was nearly identical to the HW1 multivariable-adjusted estimate (RR = 2.059, 95% CI: 1.351–3.137), demonstrating convergent validity across approaches and supporting a strong independent causal association between IVH and neonatal mortality in VLBW infants.